

Forensic Analysis Report: ICS Packet Captures Analysis

STUDENT NAME: WENDELL TWUMASI

STUDENTS ID: C11/26/ACIS/17355

Target PCAPs: ICDFA_Lab7ICS_Traffic

Date: October 1, 2026

1. Laboratory Task Evidence & Screenshots

Task 1: Basic HTTP Authentication Inspection

- **Wireshark Display Filter:** http.authbasic
- ICDFA_Lab7ICS_Traffic

SCREENSHOT 1

kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox

ICDFA_Lab7ICS_Traffic.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http.authbasic

No.	Time	Source	Destination	Protocol	Length	Info
15	0.010499	192.168.2.111	192.168.88.115	HTTP	169	GET /view
1215	0.910500	192.168.2.111	192.168.88.115	HTTP	177	[TCP Prev
2415	1.810499	192.168.2.111	192.168.88.115	HTTP	169	[TCP Prev
3615	2.710500	192.168.2.111	192.168.88.115	HTTP	177	[TCP Prev
4815	3.610500	192.168.2.111	192.168.88.115	HTTP	169	[TCP Prev

Frame 15: Packet, 169 bytes on wire (1352 bits), 169 bytes captured

Ethernet II, Src: VMware_aa:10:20 (00:50:56:aa:10:20), Dst: AxisCom

Internet Protocol Version 4, Src: 192.168.2.111, Dst: 192.168.88.11

Transmission Control Protocol, Src Port: 41007, Dst Port: 80, Seq:

Hypertext Transfer Protocol

GET /view/ HTTP/1.1\r\n

Host: 192.168.88.115\r\n

User-Agent: ICDFA-Lab-HMI/1.0\r\n

Authorization: Basic YWRtaW46YWRtaW4=\r\n

\r\n

[Response in frame: 16]

[Full request URI: http://192.168.88.115/view/]

0000 00 40 8c 12

0010 00 9b 00 0f

0020 58 73 a0 2f

0030 fa f0 04 85

0040 20 48 54 54

0050 20 31 39 32

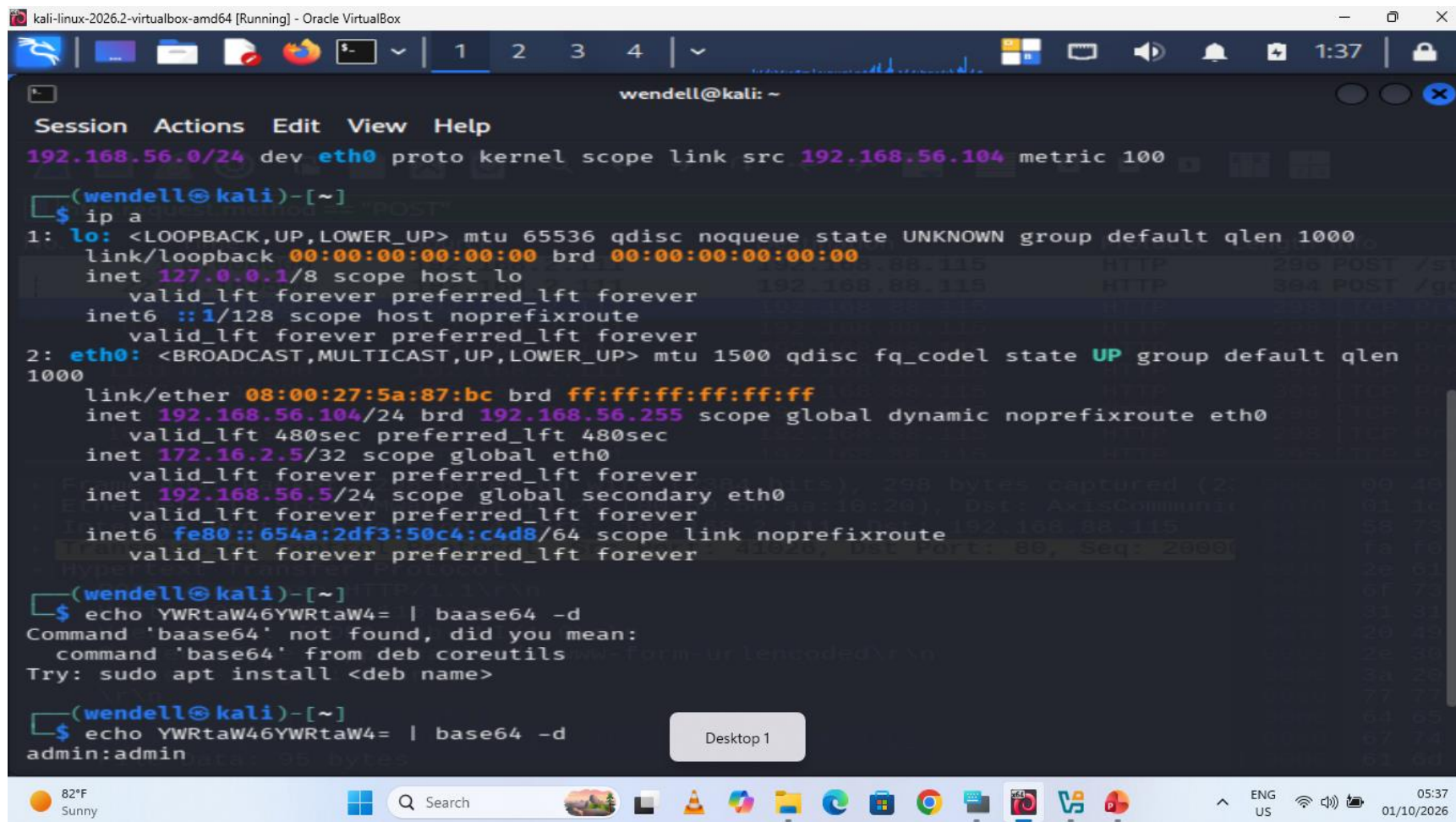
0060 0a 55 73 65

0070 46 41 2d 4c

0080 41 75 74 68

0090 61 73 69 63

00a0 74 61 57 34



Description: Wireshark displaying filter 'http.authbasic' with Packet 571931 selected, showing the expanded Hypertext Transfer Protocol pane and adjacent terminal/CyberChef window demonstrating Base64 decoding.

Task 2: HTTP Request & Form Submission Analysis

- **Wireshark Display Filter:** http.request.method == "POST" (Total filtered count: 15 packets)
- **Packet 833139 (192.168.2.111 → 192.168.88.115):**
 - userid = "admin"
 - password = "[REDACTED]" (Original: admin)
 - **Device Hardware Identification:** Destination MAC 00:40:9d:28:ee:0d maps to vendor **Digi International / Digiboard**.
- **Packet 993570 (192.168.2.88 → 192.168.88.61):**
 - Cookie String: AccountName508=admin; Password508=3f7caa3d471688b704b73e9a77b1107f
 - Hash Type identified via hash-identifier: **MD5** (Cracks to ADMIN).

SCREENSHOT NOT AVAILABLE

Description: Wireshark displaying filter 'http.request.method == "POST"' with Packet 833139 selected, showing the expanded HTML Form URL Encoded pane alongside the hash-identifier output terminal window.

Task 3: FTP Traffic & Banner Harvesting

- **Wireshark Display Filter:** ftp
- **Packet 480883 (192.168.88.49 → 192.168.2.64):**
 - **MAC Address OUI:** 00:40:8c:7f:0b:bc (AxisComm_7f:0b:bc / Axis Communications)
 - **Extracted FTP Service Banner:** 220 AXIS 206 Network Camera 4.40 (Jun 20 2006) ready.\r\n

SCREENSHOT NOT AVAILABLE

Description: Wireshark displaying filter 'ftp' with Packet 480883 selected, showing expanded Ethernet II MAC OUI details and the File Transfer Protocol (FTP) server banner.

Task 4: Modbus Protocol Inspection

- **Wireshark Display Filter:** modbus
- **4SICS-GeekLounge-151021.pcap Findings:** Modbus packets present Function Code **43** (Read Device Identification).
- **4SICS-GeekLounge-151022.pcap Findings:** Filtering yields 99,472 packets. Packet **514079** highlights Function Code 1 (Read Coils).

SCREENSHOT NOT AVAILABLE

Description: Wireshark with '4SICS-GeekLounge-151022.pcap' loaded, displaying filter 'modbus' with Packet 514079 selected, showing the expanded Modbus header with Function Code 1 (Read Coils).

Task 5: Distributed Network Protocol 3.0 (DNP3) Inspection

- **Wireshark Display Filter:** dnp3
- **Key Observations:** Captures Master and Outstation telemetry exchanges across the Data Link, Transport, and Application layers, exposing SCADA monitoring states without cryptographic verification.

SCREENSHOT NOT AVAILABLE

Description: Wireshark displaying filter 'dnp3', showcasing DNP3 Master/Outstation telemetry frames with expanded Data Link and Application layer fields.

Task 6: EtherNet/IP (CIP) Traffic Analysis

- **Wireshark Display Filter:** enip
- **Key Observations:** Details Common Industrial Protocol (CIP) encapsulated commands, session registrations, and object management communications between control endpoints.

SCREENSHOT NOT AVAILABLE

Description: Wireshark displaying filter 'enip', showing expanded EtherNet/IP and CIP header details including command codes and encapsulation headers.

Task 7: Siemens S7 Communication Inspection

- **Wireshark Display Filter:** s7comm
- **Packet 1 Findings:**
 - **Source Hardware OUI:** 00:1c:06:27:64:11 (SiemensN_27:64:11 / Siemens Numerical Control)
 - **S7 Parameter Function:** Read Var (0x04) querying 5 data variables directly from the programmable logic controller (PLC).

SCREENSHOT NOT AVAILABLE

Description: Wireshark displaying filter 's7comm' with Packet 1 selected, showing expanded Ethernet II details (Siemens MAC vendor) and the S7 Communication parameter tree for Read Var.

2. Protocol Evidence Matrix

Protocol Family	Wireshark Display Filter	Source & Destination Endpoints	Observed Operations / Extracted Data	Security Significance
HTTP	http.authbasic	192.168.2.42 → 192.168.88.25	• Base64 Auth (admin:[REDACTED])	Plaintext protocols allow passive eavesdroppers to recover valid operator credentials without

Protocol Family	Wireshark Display Filter	Source & Destination Endpoints	Observed Operations / Extracted Data	Security Significance
	http.request.method == "POST"	192.168.2.111 → 192.168.88.115 192.168.2.88 → 192.168.88.61	• Plaintext POST (admin:[REDACTED]) • Cookie Hash (MD5:[REDACTED])	executing active exploitation.
FTP	ftp	192.168.88.49 → 192.168.2.64	Banner: 220 AXIS 206 Network Camera 4.40 (Jun 20 2006) ready.	Explicit product banners inform attackers of exact device models and outdated firmware, facilitating targeted CVE searches.
Modbus	modbus	192.168.2.88 ↔ 192.168.88.x	• FC 43 (Read Device ID) • FC 1 (Read Coils)	Modbus TCP contains zero authentication or encryption, leaving register reads and coil state changes completely unprotected.

Protocol Family	Wireshark Display Filter	Source & Destination Endpoints	Observed Operations / Extracted Data	Security Significance
DNP3	dnp3	Master ↔ Outstation endpoints	Master/Outstation polling and telemetry state exchanges	Legacy DNP3 installations lack origin authentication, allowing adversaries to spoof telemetry or inject unauthorized control commands.
EtherNet/IP	enip	192.168.2.x ↔ 192.168.88.x	CIP Encapsulation headers, Session Registration, Object requests	Exposes internal industrial automation architecture, CIP object definitions, and operational states to network sniffers.
Siemens S7	s7comm	192.168.2.x ↔ 192.168.88.x	Read Var (Function 0x04) querying 5 PLC memory variables; Siemens MAC OUI (00:1c:06)	Unauthenticated PLC communications allow unauthorized network nodes to inspect or tamper with internal memory structures.

3. Knowledge Check Answers

Q1: Why does Base64 encoding not provide confidentiality for HTTP Basic authentication?

Base64 is a standard encoding scheme that can be used as a reversible encoding mechanism for binary data to ASCII string format. It does not use any secret keys or initialization vectors. The encoding process uses mostly common mathematical algorithms that are publicly known. Any individual can decode Base64 by decoding the string using readily available decoding tools.

Q2: What information can device banners or protocol metadata reveal to an analyst?

Device banners and protocol metadata contain valuable information about the target such as manufacturer, device model, operating system, application version, and timestamp of the build for certain devices (e.g., AXIS 206 Network Camera 4.40). Threat actors can use the data to find relevant common vulnerabilities and exposures (CVE) in the public exploit repository and launch an attack by compromising the firmware.

Q3: How do Modbus, DNP3, EtherNet/IP and S7 differ from ordinary user-facing protocols such as HTTP or FTP?

In contrast to the user applications, ICS protocols are directly related to the real-time control loops and field operations (i.e., PLCs, RTUs, sensors, and actuators). Moreover, many of the legacy ICS protocols were originally designed for isolated serial communications where reliability and microsecond-level response were prioritized over security. As a result, many legacy ICS protocols lack intrinsic user authentication and transport encryption mechanisms, and cryptographic message integrity checks.

Q4: Why is passive packet analysis valuable for understanding an ICS environment?

Passive packet analysis is a method of network traffic inspection, which does not involve sending or receiving any packets on the wire. Internally used embedded systems, legacy PLCs, or other operational controllers may have a sensitive or fragile network stack that may cause denial-of-service conditions and require rebooting or activation of safety mechanisms upon receiving packets from intrusive network tools (e.g., Nmap). A passive packet inspection approach guarantees zero touch and provides full visibility into the network topology and traffic content.

Q5: What compensating controls can reduce risk where legacy industrial protocols cannot be replaced immediately?

Where protocol upgrades are not feasible due to hardware lifetime constraints, organizations should deploy defense-in-depth compensating controls:

- **Purdue Model Segmentation:** Enforce strict network zoning using industrial firewalls and unidirectional data diodes between Enterprise (L4/L5) and Operations (L0–L3).
- **Encrypted Tunnels:** Enclose cleartext protocol traffic inside IPsec or TLS VPN tunnels across boundary links.
- **ICS-Aware Deep Packet Inspection (DPI):** Deploy specialized Intrusion Detection Systems (such as Malcolm, Dragos, or Snort with ICS preprocessors) configured to block write commands and alert on anomalous function codes.
- **Strict Access Control:** Enforce static ARP mapping, MAC filtering, and multi-factor jump-box access controls for administrative access.

4. One-Page ICS Security Assessment Report

To: Lead Security Engineer / Asset Owner

From: WENDELL TWUMASI

Date: October 1, 2026

Subject: Technical Forensic Evaluation — ICS PCAP Traffic Datasets

Executive Summary

A passive network traffic forensic audit was performed on the provided packet capture files (4SICS-GeekLounge-151021.pcap and 4SICS-GeekLounge-151022.pcap) representing an operational ICS environment. The assessment revealed serious OT security weaknesses at multiple levels including web management plane and core operational technology protocol layers which could be leveraged to cause operational disruptions. Exposed credentials, weak hashing algorithms, banner disclosures and lack of authentication on critical control traffic were amongst the high risk findings identified.

Key Risk Findings

1. Cleartext Administrative Credential Exposure (High Risk):

HTTP Basic authentication (http.authbasic) transmitted Base64-encoded credentials (admin:[REDACTED] and root:[REDACTED]). Web form submissions (/goform/svLogin) similarly sent unencrypted credentials over POST requests. Any node with access to internal network traffic can capture these credentials to compromise management web consoles.

2. Obsolete Cookie Hashing Mechanisms (High Risk):

HTTP POST interactions with endpoint 192.168.88.61 revealed session cookies storing user passwords as raw MD5 hashes (Password508=3f7caa3d...). MD5 lacks collision resistance and can be cracked via rainbow tables to yield underlying passwords (ADMIN).

3. Asset Infrastructure Information Disclosure (Medium Risk):

FTP service banners contain specific hardware and firmware data of the manufacturer (220 AXIS 206 Network Camera 4.40). The analysis of Mac address OUI revealed the vendor of the hardware used, including Axis Communications, Digi International, Moxa, and Siemens. It was possible for attackers to collect this information and create a list of targets to attack using specific CVEs.

4. Insecure Industrial Control Protocols (High Risk):

Modbus TCP (Function Codes 1 and 43) and DNP3 communications operate entirely in cleartext without origin authentication or frame integrity validation. An adversary on the local network segment can issue unauthorized control function codes or manipulate field registers.

5. Unauthenticated Siemens S7 PLC Variable Polling (High Risk):

Siemens S7 (s7comm) traffic demonstrated active Read Var operations (0x04) pulling variable states directly from PLCs without session authentication. In an unsegmented network, malicious actors could issue unauthorized write commands to modify PLC logic.